

The Virtual Memory Palace as an Interactive Mnemonic Learning Platform

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Abstract—We introduce a demo application of a memory palace in virtual reality (VR) and describe findings from an initial usability and observational pilot evaluation with six users. We further identify the conceptual elements of memory palaces, based on ancient guidelines and in relation to prior literature, and show how we used them to shape the design and experience of our application, specifically its interfaces and interactions.

Index Terms—virtual reality, memory palace, usability, study

I. INTRODUCTION

Learning and memorising vocabulary or historical events has never been easy and people often struggle to effectively memorise, retain, and recall as we often also neglect the power of context. The memory palace (also called method of loci, journey method, or memory theatre) is a learning and memorising technique that actively taps into these neglected areas by engaging with the contextual structure of the content. Memory palaces are known since the ancient times and have moved through history. In the middle ages the philosopher Giulio Camillo [1] and Jesuits like Matteo Ricci [2] practised memory palaces. Related methods were studied in classroom settings over the past few decades [3], [4] that highlighting its benefits. The method basically relates what we want to learn (e.g. a list of names) with an interesting story and a real or imagined location (e.g. the corners of our room or the chambers of our imagined 'palace'). When we want to learn a list of names, we first reorganise them into a connected story of mnemonics, or images, that we distribute along a pathway in the space of the memory palace. Building and recalling a sequence of contextually connected story elements along a well-known (real or imagined) place is easier for our brain to process. Creating and recalling now follows the contextual story instead of a flat list of terms.

This paper introduces a demo application, described alongside key concepts of memory palaces and how they shaped its design. We present results from a first usability and observational pilot study that will help to extend the application into an experimental platform for user studies in this domain.

II. RELATED WORK

The success of VR in the past decade has been triggered by an array of consumer head-mounted displays and currently

has over 4000 exclusive VR software products (mostly games). Educational applications are often viewed as another promising future domain [5]. The ancient method of the memory palace may belong to this educational category. It originated in the Greek and the Roman civilisation as a tool for memorising speeches and lectures, it was handed down through the Middle Ages to the modern times [1], [2]. Besides some interest in education that studied the technique in classroom settings [3], [4], [6], memory palaces have unfortunately now been mostly reduced to an artifact of pop-culture with some limited interest for VR. As an early example, [7] applied the method to transform the fully mental activity into a computer-aided format. Böerner in [8] explored memory palaces as a shared 3D browsing activity and through a so-called CAVE — a room-size installation with projectors [9]. Later examples, such as Fassbender and Helden [10], allowed users already to freely explore a (virtual) 3D-scene (without the use of a VR headset) to place icons in key locations. Huttner and Robra-Bissantz [11] investigated the technique comparatively between screen and VR and found that the performance with VR was 5-7% higher. Likewise, Krokos and colleagues [12] similarly discovered that memory recall improved at about the same rate (9%) by arranging pictures spatially. While Huttner [11] let participants traverse their memory place, [12] only allowed for passive viewing of mnemonics. Huttner et. al [13] later presented learning content either in textual or in mixed (visual and text) format without interactive features. Legge et. al. [14] also excluded interactive features with mnemonics as an option. All four studies did therefore not allow users to actively create mnemonics in the virtual space nor did they allow users to interact with them. Such abilities, however, could be essential for harvesting the full potential of the memory method. In this work, therefore, we will demonstrate a memory palace implementation that offers such capabilities. We describe its features alongside the aspects of the space, the learning topic, and the creation and interaction with mnemonics. We also report on an initial user evaluation.

III. MEMORY PALACE DEMO

The demo implements three basic aspects that are needed for creating and exercising learning scenarios with memory palaces — an ordered *space* (or loci [1]), a feature for

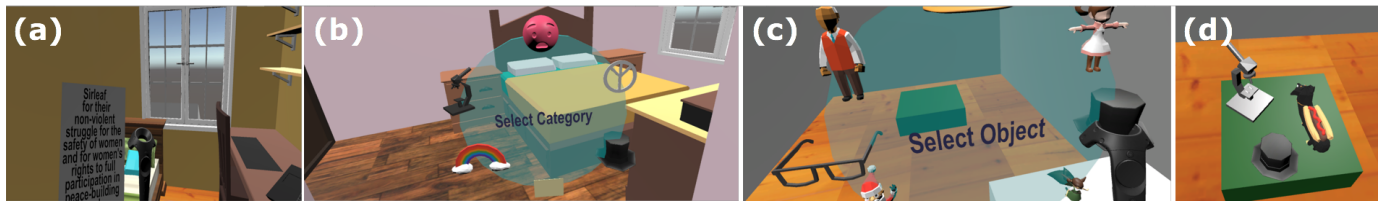


Fig. 1. The memory palace demo application is implemented as a house with multiple rooms. Participants were guided by instructions throughout different rooms (or loci) (a), selected suitable objects from a menu divided by high-level object categories (c) that led to specific objects choices (b), and placed these in different rooms (or loci) (d) to form their personalised mnemonics.

providing a *topic* to a learner, and interactive features for generating vivid *stories* through access to pictorial 3D-objects.

A. Space for Structure

The space of the memory palace should follow some fundamental guidelines as summarized in [1] and should be:

- *Imaginable (and familiar)*: Loci can be entirely imaginary, as long as a user can accurately imagine it. This leads to an implicit requirement of familiarity with the space. It represents a challenge since users are new to the space and are expected to navigate it rather than imagining it. According to the 2016 Irish census [15] 70% of households live in detached and semi-detached houses as the most common type of accommodation in Ireland. We therefore chose a semi-detached three-bedroom five-person house as the basis for our design. Most people are familiar with typical rooms that can be found inside such a house: kitchen, living room, bathroom, bedrooms, and hallways connecting them. We used a representative floor plan for the Irish market from the Centre for Excellence in Universal Design [16] and applied ideas for typical furniture and decorations based on photographs of furnished rooms from Irish letting agencies (for example, see fig. 1(a) and (b)).
- *Ordered*: The space should consist of an ordered series of places (loci) that, once determined, should not be altered. Users can then start anywhere and traverse it bidirectionally. The house and its potential loci were presented as a series of rooms in standard spacial arrangement. Here, users can find their own personal singular path through the house (e.g. going to the kitchen before retreating to the home office) or using its natural order (e.g. kitchen and living room being connected).
- *Emptied*: When a new set of loci is recorded, it should be empty of people as this might distract from the images the user creates. By design the house was empty of people and this guideline also suggests that there will be no need for a collaborative feature in the future.
- *Distinct*: Users should be able to differentiate each locus from any other. The rooms of the house naturally offer different meaning and function (e.g. a kitchen for high activity and cooking, a bedroom for relaxation and rest).
- *Accessible*: The locus should be well lit, of moderate size and the loci should be a moderate distance apart, so that the image assigned to them is clear and separate

from other images. By design the house is sufficiently large and bright with distinctive rooms and enough space between furniture that might serve as loci.

- *Segmented*: Every fifth locus should have a special marker to help the user remember where they are in the series. This represents a challenge as the space is not completely visible and hence it is difficult to implement markers in a straightforward and universal way. However, the space also has partial orders that can be used for spatial segmentation, e.g. bedrooms being close.

In addition to these features, we also chose an overall low-poly (or caricaturist) graphical style [17, p. 143] for the house, including its furniture, its objects and the interfaces. Research has shown that low-poly graphics often do create the same effects than higher-quality graphics in VR and gaming settings, e.g. for memory recall [14], relaxation [18], or immersion [17].

B. Learning Topic

We chose winners of the Nobel Prize — a prestigious award that documents some of the most important breakthroughs in human history — as our example learning topic provided by the Nobel Foundation as a Kaggle data set¹. While the Nobel Prize is widely known, few people have comprehensive historical knowledge of complete sets of exact names and their achievements. This means that while the topic requires little explanation, it promises a useful, motivating, and engaging subject. The Nobel Prize is intrinsically chronological and awarded annually providing the name of the winner (word) and the description of the achievement (concept or event). Throughout history, the memory palace has been used to memorise these two types of knowledge — literate wordings (e.g. for a speech) and conceptual aspects (e.g. facts and relationships of a court case) [1, p.7-8]. We selected two Nobel categories (physics and peace) with seven winners and their achievements from each. Winners had to be lesser-known individuals (e.g. not Barack Obama) with their achievement expressed in an official statement. Based on our own judgement, we selected these statements of achievement so that they were comparable in their difficulty and also comprehensible for an average user without deeper background in the given field. For physics, for example, we preferred '*James Chadwick for the discovery of the neutron*' rather than '*William Alfred*

¹<https://www.kaggle.com/nobelfoundation/nobel-laureates>

Fowler, for his theoretical and experimental studies of the nuclear reactions of importance in the formation of the chemical elements in the universe’ for being shorter and students being generally familiar with neutrons from Junior Certificate level.

Users consume their task information and the details about their seven Nobel’s via a simple tablet text display attached to the left (or the non-active handed) HTC controller that is always available (figure 1(a)). This allows users to intuitively adjust its position to, for example, bring the text closer to the user’s eyes or to move it out of one’s view.

C. Creating Stories from Pictorial Objects

Our implementation allows users to actively create mnemonic images from a given set of objects provided by a menu. Such mnemonic images allow users of memory palaces to encode items from the learning topic by placing multiple objects together for easy recall. For example, to represent the name ‘Sirleaf’, one of the Nobel Prize winners, users might pick a top hat (to represent ‘Sir’) and add a leaf object. The object menu holds about fifty objects which is comparable to standard inventory systems that are often found in adventure and role-playing games. This interactive feature exceeds other implementations that asked users to only imagine objects [14], place simple pictures [10], or view them in fixed locations [12]. We populated the menu with manually selected objects that related to the subject matter of the Nobel category described before (physics or peace). The menu has two levels — category level (e.g. technology) and object level (e.g. microscope) — for fast access and easy use. We applied a low-poly (or caricaturist) graphical style [17, p. 143] on all objects (including the menu interface) to make them usable in a wide range of contexts. A top hat, for example, can represent all types of hats instead of a specific hat worn by a particular person. Similarly, emoticons represent a reliable shorthand to represent emotions. This allowed us to maximise the expressive possibilities from a reduced set of about 50 objects. All objects are fully interactive to fulfil the guidelines of the memory palace. Specifically, they can be selected from the menu, placed freely anywhere in the house, picked up again and rearranged, and deleted.

IV. INITIAL EVALUATION

We evaluated the virtual memory palace with six volunteering participants as part of a preliminary usability and observational pilot study with the goal to a) evaluate the usability of the virtual environment of the memory palace b) evaluate the menu system for choosing and interacting with reference objects, and c) collect initial qualitative feedback from users performing a specific learning task. We installed an interaction laboratory at our university using a HTC Vive VR headset with two controllers in a room-scale configuration. Consenting participants were brought to the laboratory. A separate training room, part of the virtual memory palace application, was used to get familiar with specific features, such as moving and teleporting, reading instructions, using the object menu, and interacting with objects. Participants were asked to follow the

text instructions that were shown on the left controller as a floating information tablet (see figure 1(a)). Participants were allowed into the memory palace thereafter and asked to create mnemonics to memorise seven Nobel Price winners as a task. They were also encouraged to navigate through all the rooms of the house. The total session time (per participant) was between 20-30 minutes on average. Participants’ expressions and remarks were documented by the researcher during the session. After the session, participants completed a standard usability questionnaire (SUS) [19] and answered questions about any negative, interesting, and positive experiences in free-form. After the conclusion of the session, participants were released.

A. Findings

We used the System Usability Scale [19] — a common usability measure that is also widely used for evaluations in VR. Participants completed this questionnaire about two system aspects: a) the holistic memory palace interface and experience and b) the object menu that participants used to create mnemonics. Additionally, they provided information about any negative, interesting, and positive experiences in free form. During the study, the researcher also recorded specific observations from participants that we report separately. We used the memory task to make users behave naturally and therefore do not elaborate on the learning aspect.

1) *Usability Questionnaire*: We collected the usability questionnaire from five participants — one participant accidentally observed another participant during a session and was therefore removed from the SUS data. The memory palace SUS score was 75.5; the menu SUS score was 82.5. With both scores being well above the SUS average of 68, this suggests that participants found the memory palace generally usable.

2) *Observations*: Participants naturally expressed their thoughts during the session that were documented that gave us some initial and qualitative feedback.

- *Interaction and Fun*: Many attempted to interact with the objects from the menu similar to the real world (e.g. throwing or stacking objects). One participant tried to wear a pair of virtual glasses; another tried to put a hat object on top of a man object. These behaviours suggest that people seek interaction and try to discover the interactive affordances of objects available to them or attempt to compose objects into arrangements. In future versions, we will add features that enable object interactions and compositions for more dynamic storytelling.
- *Engagement with Obscurity*: Many participants selected the Wiener Dog and the Toilet objects from the menu when they first saw them. None of the participants used these objects as part of their mnemonic arrangements, however, the general popularity of these objects suggest that unusual or out-of-place objects may be as memorable in VR as they were in ancient memory palaces [1], [2]. In future versions of the application, we will therefore consider collections of obscure objects and interactions that allow users to intensify their mental imagery.

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